P15

The Fourier reconstruction of a head section

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Abstract

The Fourier reconstruction may be viewed simply in the spatial domain as the sum of each line integral times a weighting function of the distance from the line to the point of reconstruction. A modified weighting function simultaneously achieves accuracy, simplicity, low computation time, as well as low sensitivity to noise. Using a simulated phantom, the authors compare the Fourier algorithm and a search algorithm. The search algorithm required 12 iterations to obtain a reconstruction of accuracy and resolution comparable to that of the Fourier reconstruction, and was more sensitive to noise. To speed the search algorithm by using fewer interactions leaves decreased resolution in the region just inside the skull which could mask a subdural hematoma.

Real-World Impact & Application

How Hospital CT Scanners See Inside Your Brain: A CT scanner takes 2D X-ray slices around your head and uses math to rebuild a 3D picture of your brain. This paper created the algorithms and the famous 'Shepp-Logan Head' test model that every hospital scanner in the world still uses to calibrate before scanning patients.